Assignment-2.

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Task-1: Linear Regression.

40/50

$x = 1, 2, 3, 4, 5$

$y = 2.1, 3.9, 6.2, 7.8, 10.1$

a) w_0, w_1

$\bar{x} = 3, \quad \bar{y} = 6.02$

$w_1 = \frac{S_{xy}}{S_{xx}} = 19.9/10 = 1.99$

$w_0 = \bar{y} - w_1 \bar{x}_1 = 6.02 - 5.97 = 0.05$

$y = 0.05 + 1.99x$

b) predict $x = 6$.

$y = 0.05 + 1.99(6) = 11.99.$

5

c) RSS, MSE, R^2

$RSS = \Sigma(y - \hat{y})^2 = 0.107$

$MSE = RSS/n = \frac{0.107}{5} = 0.0214$

$R^2 = 1 - RSS/SST = \frac{1 - 0.107}{39.708} = 0.9973$

d] The prediction at $x = 6$, is reasonably reliable because the data can be strong.

(b) Task-2: Ridge Regression.

$S_{xx} = 10, \quad S_{xy} = 19.9, \quad \lambda = 5$

$$W_{ridge} = \frac{S_{xy}}{S_{xx} + \lambda} = \frac{19.9}{15} = 1.3267$$

$$W_{LS} = 19.9/10 = 1.99$$

2

percentage shrinkage

$$1.99 - 1.3267 / 1.99 \times 100 = 33.33.$$

Task-3: Naive Bayes.

P (spam) = P(Ham) = 0.5

		Spam	Ham
offer	= y	0.8	0.2
Link	= y	0.8	0.2

4

For (Y, Y) without Smoothing P(spam) = 0.9412.

with Smoothing P(spam) = 0.8621.

For (N, N).

With Smoothing : P(Spam) = 0.1379 → Ham.

- Smoothing avoids zeros probabilities.

Task-4: Logistic Regression.

Intial probability = 0.5

2

Gradient (0.25, -0.25)

gradient = 0

$w = (-0.125, 0.125)$, $b = 0$.

Predictions : 0.5312, 0.4688.

Intial Loss = 0.6931.

Task-5: Decision Tree.

a) parent entropy

$$H(S) = -0.6\log_2 0.6 - 0.4\log_2 0.4$$

$$H(S) = 0.971.$$ ✓

b) $H(S/A) = 0.4(0) + 0.6(0.918)$

$$= 0.551.$$

c) $IG(A) = 0.971 - 0.551$ ✓

$$= 0.420 \text{ bits.}$$

3

d) Ginni decrease = 0.213. ✓

Task-6: Decision Tree pruning:

a) Cost comparision

$$C_{subtree} = 3 + 4\alpha$$

$$C_{leaf} = 8 + \alpha$$

$$3 + 4\alpha = 8 + \alpha$$

$$3\alpha = 5$$

$$\alpha = 5/3 = 1.667.$$ ✓

b) pruning decision

For $\alpha = 0.01$, $\alpha = 0.02$

both are less than 1.667. ✗ (2)

Task-7: Neural Network - Back propagation.

a) Forward propagation.

$$net_h = 0.4(1) + 0.3(2) = 1$$

$$h = \sigma(1) \Rightarrow 0.7311$$

$$net_o = 0.5(0.7311) = 0.3655$$

$$\hat{y} = \sigma(0.3655) = 0.5904.$$ ✗

b) Initial error:

$$E = 1/2 (1 - 0.5904)^2$$

$$= 0.0839.$$ ✓

c) Updated weight:

$$\delta_o = (1 - \hat{y})(\hat{y}(1 - \hat{y})) \approx 0.09$$

$$v = 0.5363.$$ ✓

d) Updated error: (2)

$$E_{new} = 0.0811.$$ ✗

Task-8: Support vector machine.

$f(x) = 0.5x_1 + 0.5x_2 - 2$

$(3,3)(1,1) - \|w\| = 0.7071$ ✓

Width = 2.828.

Hinge loses:

$(1,2) = 0.5$

$(2,2) = 1$ ✓

(3)

Task-9: Ensemble.

a) For 3 classifiers → 78.4 %. ✓

b) For 5 : 83.69. ✓

(3)

Improvement = 5.292.

Carrrelated errors due to ensemble improvement.

Task-10: Ada Boost.

Weighted error: $E = 2/5$

$= 0.4$ ✓

Learner weight:

$\alpha = 1/2 \ln(1.5)$

$= 0.2027.$ ✓

Normalized weights:

misclassified : 0.25 each

correct : 0.1667 each ✓ (3)

Task-11:

$w = (10/7, -12/7)$

Equal prior Threshold

$\Rightarrow 2/7$. (2)

For $x = (2.5, 1)$: class ω_2

with unequal prior threshold $\simeq 1.672$. ✓

Task-12:

Metric	A	B	C
Accuracy	88%	80.5%	93% ✓
precision	43.75%	32.08%	66.67%
Recall	70%	85%	60%
F1	53.85%	46.58%	63.16%
Cost	₹ 19.5 Lakh	₹ 16.5L	₹ 21.5L ✓

(1)

Best model : B.

It has Lowest cost & highest recall. ✓